

DSA Bootcamp - Day 1 Revision (Binary Search Foundations)

Core Concepts

1. Binary Search maximizes information gained per comparison.
2. Middle minimizes the worst-case remaining search space.
3. Think in terms of search interval, not just array.
4. Ordering guarantees safe elimination.
5. Invariant: if target exists, it stays inside [low, high].
6. Interval shrinks every iteration, so the algorithm terminates.
7. Use $\text{mid} = \text{low} + (\text{high} - \text{low}) / 2$ to avoid overflow.
8. $\text{low} = \text{mid}$ may cause infinite loops; use $\text{low} = \text{mid} + 1$ when appropriate.
9. Binary Search also works on monotonic answer spaces.
10. Infinite-array approach: exponential expansion + Binary Search.

Interview Keywords

Search Space, Ordering Guarantee, Information Gain, Loop Invariant, Monotonic Property, Monotonic Predicate, Divide and Conquer, Random Access, Overflow Prevention, Feasibility Check.

Interview Review

Q1 Dictionary: You focused on character comparison. Improve by explaining elimination of half the search space.

Q2 Guess Number: Good reasoning. Use the phrase 'reduce the search space by half'.

Q3 Unsorted Array: Good. State that elimination is no longer justified mathematically.

Q4 Linked List: Correct. Mention lack of random access.

Q5 Expensive Comparisons: Correct intuition. Quantify $O(n)$ vs $O(\log n)$.

Q6 Termination: Mention monotonically shrinking finite interval.

Q7 Infinite Loop: Correctly identified $\text{low} = \text{mid}$ issue.

Q8 Infinite Array: Correct direction after discussion: expand exponentially, then binary search within the discovered interval.

Golden Statements

- Binary Search is about proving where the answer cannot be.
- Algorithms are derived from properties, not memorized.
- Good algorithms maximize information gained per operation.
- The deepest requirement is the ability to eliminate part of the search space with certainty.

Tomorrow

Loop invariant → derive code → dry run → first 5 implementation problems → interview round.